[인공지능]

Clustering

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Example of Machine Learning Tasks

Task	Input	Target	Model
Car price prediction	Age, mileage, brand	Car price	Linear regression
Electricity forecasting	Weather, time, past usage	Next-hour usage	Linear regression
Customer churn	Usage, fees, complaints	Churn / Stay	Logistic regression, SVM
Fault diagnosis	Vibration, temperature, current	Normal / Fault type	Multi-class SVM
Activity recognition	Accelerometer, gyroscope features	Walking / Sitting / etc.	Multi-class SVM

Unsupervised Learning

- Learning patterns and structures from unlabeled data.
  - No ground truth labels are provided.
  - The model discovers hidden structures or relationships.

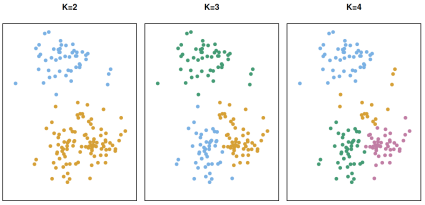
Clustering

- **Clustering** refers to a broad set of techniques for finding subgroups, or clusters, in a data set.
- We seek a partition of the data into distinct groups so that the observations within each group are quite similar to each other.
- We must define what it means for two or more observations to be similar or different.
- Indeed, this is often a domain-specific consideration that must be made based on knowledge of the data being studied.

Clustering Methods

- In **K-means clustering**, we seek to partition the observations into a pre-specified number of clusters.
- In **hierarchical clustering**, we do not know in advance how many clusters we want; in fact, we end up with a tree-like visual representation of the observations, called a dendrogram, that allows us to view at once the clusterings obtained for each possible number of clusters, from 1 to n.
- In **DBSCAN**, we identify clusters as dense regions of points separated by areas of lower density, without requiring the number of clusters in advance.

K-Means Clustering



A simulated data set with 150 observations in 2-dimensional space

- Panels show the results of applying K-means clustering with different values of K, the number of clusters.
- The color of each observation indicates the cluster to which it was assigned using the K-means clustering algorithm.
- Note that there is no ordering of the clusters, so the cluster coloring is arbitrary.
- These cluster labels were not used in clustering; instead, they are the outputs of the clustering procedure.

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Details of K-means Clustering

- Let $C_1, \ldots, C_K$ denote sets containing the indices of the observations in each cluster.
- These sets satisfy two properties:
  1. $C_1 \cup C_2 \cup \cdots \cup C_K = \{1, \ldots, n\}$.
     In other words, each observation belongs to at least one of the K clusters.
  2. $C_k \cap C_{k'} = \varnothing$ for all $k \neq k'$.
     In other words, the clusters are non-overlapping: no observation belongs to more than one cluster.
- For instance, if the ith observation is in the kth cluster, then $i \in C_k$.

- The idea behind K-means clustering is that a **good clustering** is one for which the within-cluster variation is as small as possible.
- The within-cluster variation for cluster C_k is a measure $\text{WCV}(C_k)$ of the amount by which the observations within a cluster differ from each other.

$$\min_{C_1,\ldots,C_K} \left\{ \sum_{k=1}^{K} \text{WCV}(C_k) \right\}.$$

How to Define Within-Cluster Variation?

- Typically we use Euclidean distance

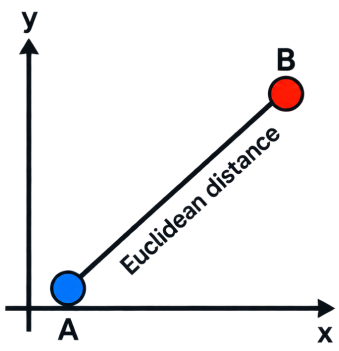
$$\text{WCV}(C_k) = \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2,$$

where $|C_k|$ denotes the number of observations in the kth cluster, and p is the number of features (dimensions).

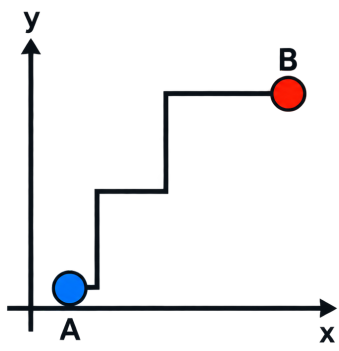
- Combining the previous definitions gives the optimization problem that defines K-means clustering:

$$\min_{C_1,\ldots,C_K} \left\{ \sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2 \right\}.$$

Distance Metrics



Euclidean Distance

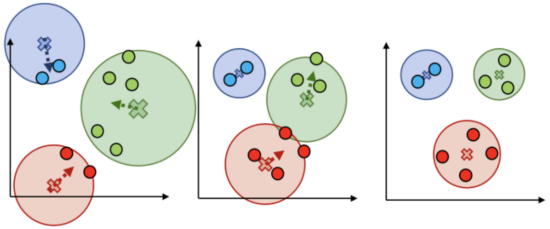


Manhattan Distance

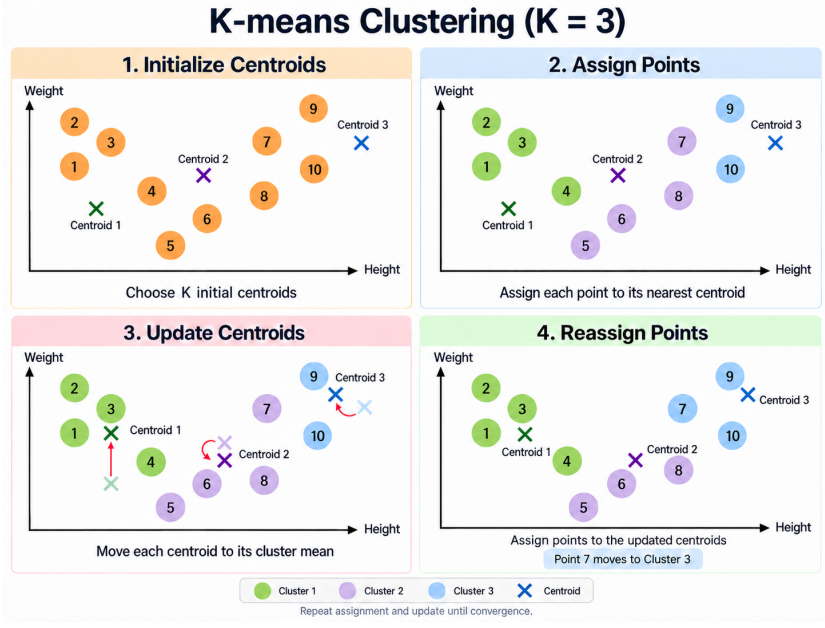
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K-Means Clustering Algorithm

1. Randomly assign a number, from 1 to K, to each of the observations. These serve as initial cluster assignments for the observations.
2. Iterate until the cluster assignments stop changing:
   - 2.1 For each of the K clusters, compute the cluster *centroid*. The kth cluster centroid is the vector of the p feature means for the observations in the kth cluster.
   - 2.2 Assign each observation to the cluster whose centroid is *closest* (where *closest* is defined using Euclidean distance).



Source from Water Programming


K-means Clustering (K = 3)
1. Initialize Centroids
Weight
Height
Centroid 1
Centroid 2
Centroid 3
Choose K initial centroids
2. Assign Points
Assign each point to its nearest centroid
3. Update Centroids
Move each centroid to its cluster mean
4. Reassign Points
Assign points to the updated centroids
Point 7 moves to Cluster 3
Cluster 1
Cluster 2
Cluster 3
Centroid
Repeat assignment and update until convergence.

K-Means++

- Problem with random initialization: Poor starting points $\Rightarrow$ convergence to poor local minima.
- Idea of K-means++: Choose initial centroids in a smarter way.
- Algorithm:
  1. Choose the first centroid uniformly at random from the data points.
  2. For each subsequent centroid, select a data point with probability proportional to its squared distance from the nearest chosen centroid.

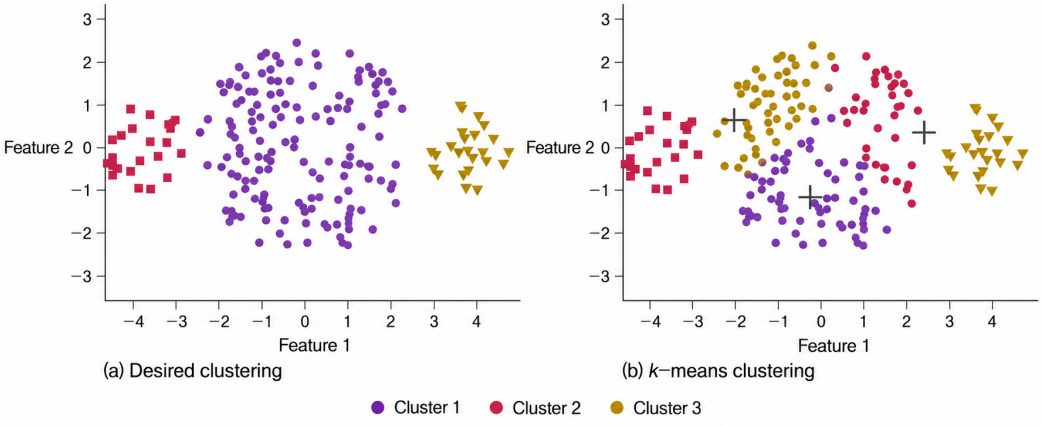
  $\rightarrow$ Points that are farther from the already chosen centroids are more likely to be selected, while closer points have a lower chance.

  3. Repeat until K centroids are chosen.
- Advantages:
  - Provides better spread-out initial centroids.
  - Often leads to faster convergence.
  - Reduces the chance of poor local minima.

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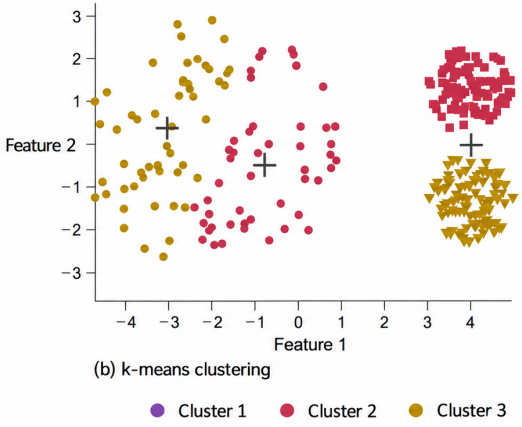
Limitations of K-Means

- Struggles to identify clusters of different sizes.

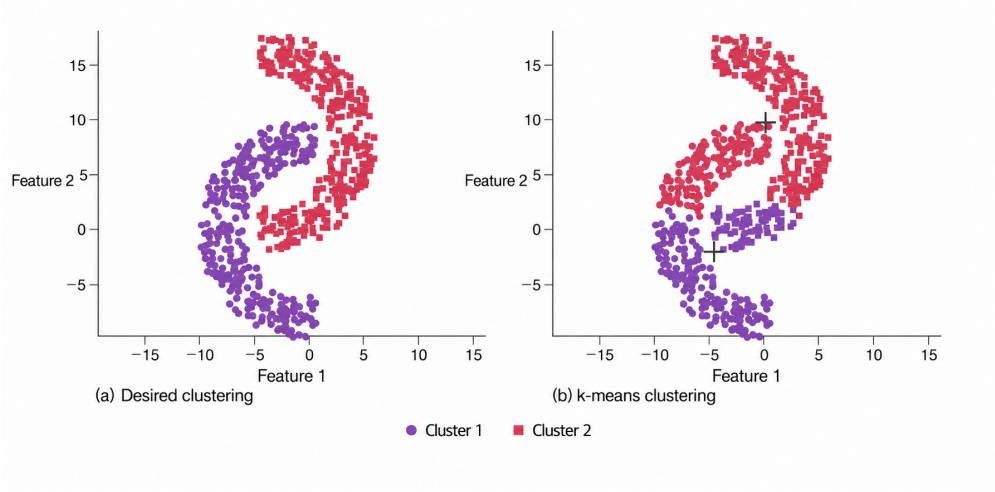


- Struggles to identify clusters of different densities.

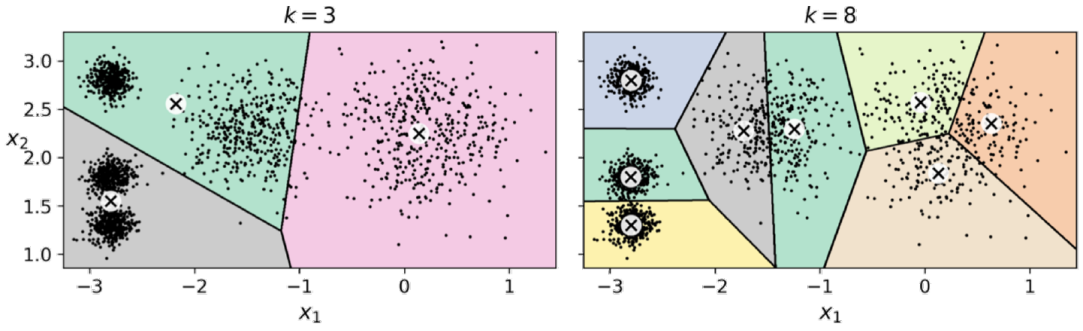




- Performs poorly when local patterns are present.



- Not be so easy to know how to set K.

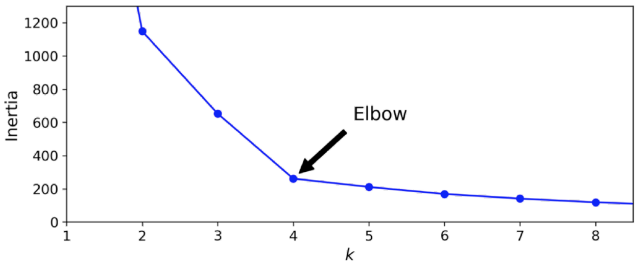


Finding the Optimal Number of Clusters 1: Inertia

- ELBOW with inertia: The inertia is not a good performance metric when trying to choose k since it keeps getting lower as we increase k.

$$\text{Inertia} = \sum_{k=1}^{K} \text{WCV}(C_k) = \sum_{k=1}^{K} \sum_{i \in C_k} \sum_{j=1}^{p} (x_{ij} - \bar{x}_{kj})^2$$

  - $\bar{x}_{kj}$: the j-th feature value of the centroid of cluster k



Finding the Optimal Number of Clusters 2: Silhouette Score

- A more precise approach for choosing k is to use the **silhouette score**, which is the mean silhouette coefficient over all instances.
- For an instance, the silhouette coefficient is defined as

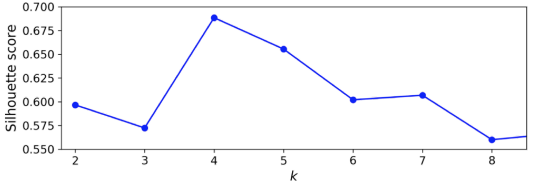
$$s = \frac{b - a}{\max(a, b)},$$

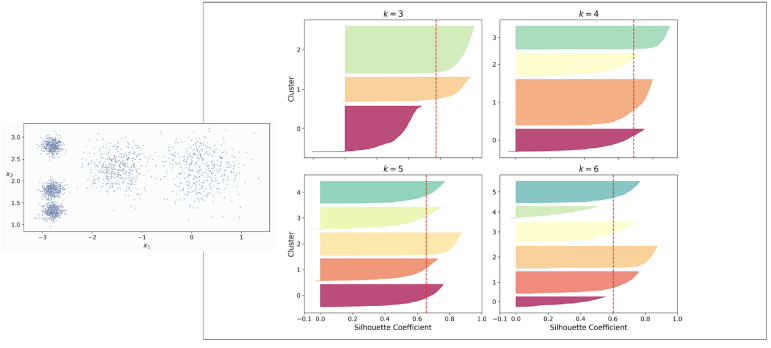
where

  - a: mean distance to other points in the same cluster (intra-cluster distance),
  - b: mean distance to points in the nearest other cluster.

  - $s \approx +1$: well inside its own cluster, far from others.
  - $s \approx 0$: close to a cluster boundary.
  - $s \approx -1$: possibly misclassified, closer to another cluster.

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Silhouette score
0.700
0.675
0.650
0.625
0.600
0.575
0.550
2
3
4
5
6
7
8
k

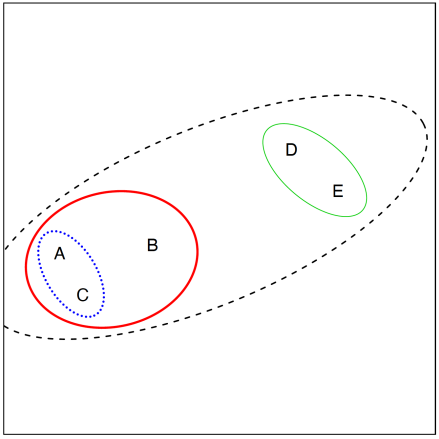

k = 3
k = 4
k = 5
k = 6
Cluster
Silhouette Coefficient
x2
x1

Hierarchical Clustering

- K-means clustering requires us to pre-specify the number of clusters K. This can be a disadvantage.
- **Hierarchical clustering** is an alternative approach which does not require that we commit to a particular choice of K.
- In this section, we describe bottom-up or **agglomerative clustering**. This is the most common type of hierarchical clustering, and refers to the fact that a dendrogram is built starting from the leaves and combining clusters up to the trunk.

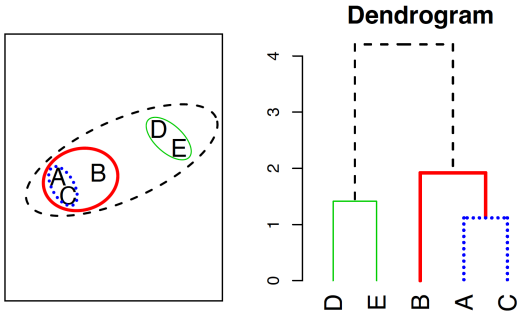
Idea of Hierarchical Clustering

- Builds a hierarchy in a bottom-up fashion.

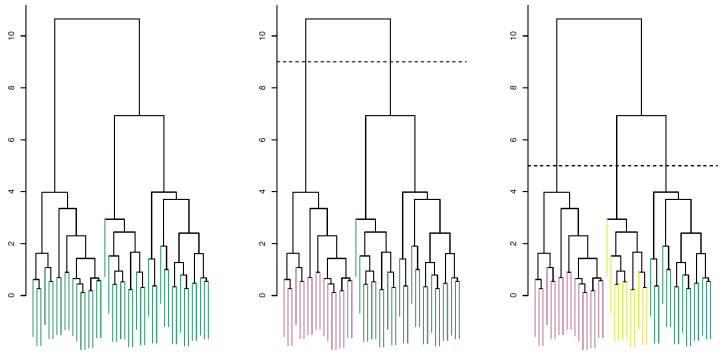


Hierarchical Clustering Algorithm

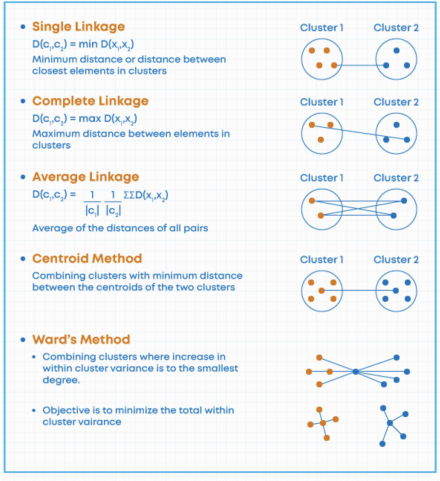
1. Start with each point in its own cluster.
2. Identify the closest two clusters and merge them.
3. Repeat.
4. Ends when all points are in a single cluster.



Application of Hierarchical Clustering



Types of Linkage



Source from DataAspirant

Density-Based Spatial Clustering of Application with Noise

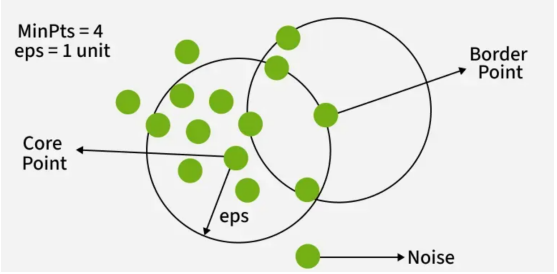
- **Density-based clustering** conduct a clustering by considering the density of data points.
  - **Density-Based Spatial Clustering of Application with Noise (DBSCAN)**
- Groups together points that are closely packed (high-density regions).
- Marks points in low-density regions as outliers (noise).
- Unlike K-means:
  - No need to pre-specify the number of clusters.
  - Can discover clusters of arbitrary shape.

DBSCAN: ε-neighborhood of a point

- Definition 1: ε-neighborhood of a point $p \in D$:

$$N_\varepsilon(p) \;=\; \{\, q \in D \mid \mathrm{dist}(p,q) \leq \varepsilon \,\}.$$

- Naïve approach: require for each point in a cluster that there are at least a minimum number (**MinPts**) of points in an ε-neighborhood of that point.
- Problem of naïve approach: An ε-neighborhood of a border point contains significantly less points than an ε-neighborhood of a core point.



- A point p is a **core point** if at least *MinPts* points (including p itself) are within its ε-neighborhood:

$$|N_\varepsilon(p)| \geq \textit{MinPts}$$

Core points are the interior of dense regions.

- A point p is a **border point** if

$$|N_\varepsilon(p)| < \textit{MinPts}$$

but p lies within the ε-neighborhood of a *core point*.
Border points are on the edge of a cluster.

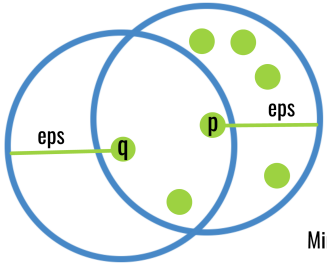
DBSCAN: Directly Density-Reachable

- Definition 2: A point q is **directly density-reachable** from a point p with regard to the parameter ε and MinPts, if

$$q \in N_\varepsilon(p) \quad \text{(reachability)}$$

$$|N_\varepsilon(p)| \geq \text{MinPts} \quad \text{(core point condition)}$$

  Property
  - Directly density-reachable is symmetric for pairs of core points.
  - Not symmetric if one core point and one border point are involved.



Directly density reachable

DBSCAN: Density-Reachable & Density-Connected

- Definition 3: A point q is **density-reachable** from p if there exists a sequence

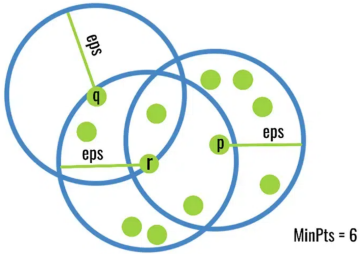
$$p = p_1, p_2, \ldots, p_m = q$$

such that for all $i = 1, \ldots, m-1$,

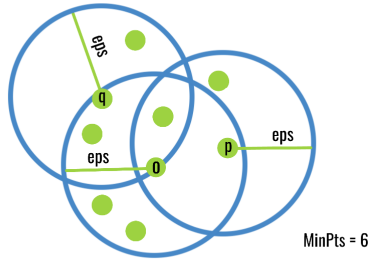
$$p_{i+1} \in N_\varepsilon(p_i) \quad \text{and} \quad p_i \text{ is a core point.}$$

- Definition 4: Points p and q are **density-connected** if there exists some $o \in D$ such that

$$p \text{ is density-reachable from } o \quad \text{and} \quad q \text{ is density-reachable from } o.$$



Density reachable



Density connectivity

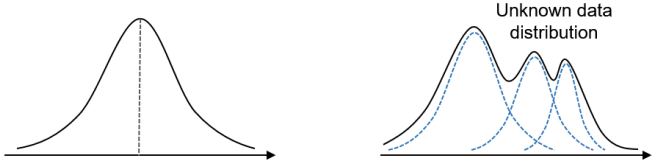
DBSCAN: Cluster & Noise

- Definition 5: A **cluster** with regard to the parameters ε and *MinPts* is a non-empty subset C of the database D with:
  1. For all $p, q \in D$: If $p \in C$ and q is density-reachable from p, then $q \in C$ (Maximality)
  2. For all $p, q \in C$: p is density-connected to q (Connectivity)

- Definition 6: **Noise**
  - Let $C_1, \ldots, C_k$ be the clusters of the database D with respect to ε and *MinPts*.
  - The set of points in D not belonging to any cluster $C_1, \ldots, C_k$ is called **noise**.

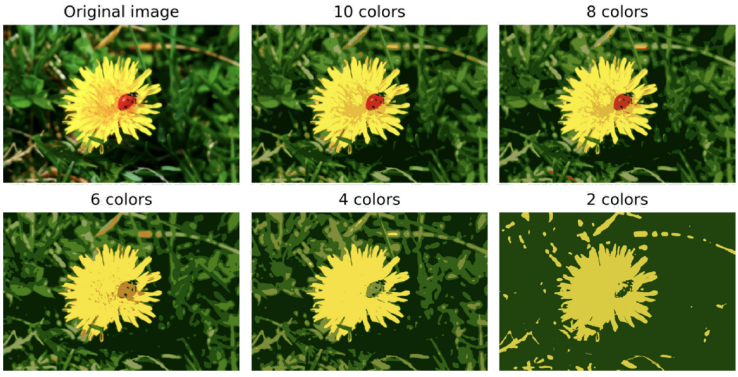
Gaussian Mixture Model

- Gaussian Mixture Model (GMM): can be understood simply as a number of Gaussians introduced into a population of data in order to classify each of the possible sample clusters, each of which could refer to classes.

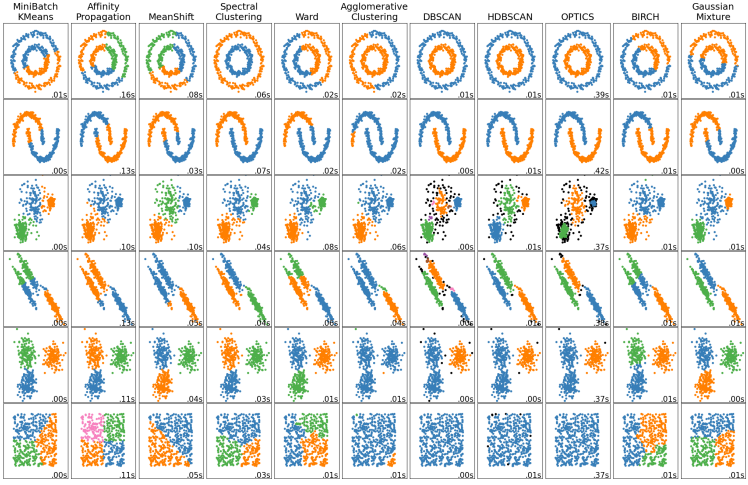


Using Clustering for Image Segmentation

- With K-Means



Other Clustering Methods



Thank You!

(Q & A)

References

- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An Introduction to Statistical Learning: with Applications in R. New York: Springer.
- Géron, A. (2019). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (2nd ed.). O'Reilly Media, Inc.